1) Discuss each of the following terms:

a) Data
Data are raw facts and figures associated with real-world objects or events. From the end-users' point of view, data constitute the values connected with entities of concern, and by themselves have little meaning until processed.

b) Database
A database is a shared collection of logically related data, together with a description of this data, designed to meet the information needs of an organization.

c) Database management system
A DBMS is a software system that enables users to define, create, maintain, and control access to the database.

d) Database application program
A database application program is a computer program that interacts with the database by issuing appropriate requests, typically SQL statements, to the DBMS.

e) Data independence
Data independence is the separation of underlying file structures from the programs that operate on them, also referred to as program-data independence.

f) Security
Security is the protection of the database from unauthorized access, often implemented using passwords, access restrictions, and authorization mechanisms.

g) Integrity
Integrity refers to the maintenance of the validity and consistency of the database by enforcing constraints on the data.

h) Views
Views present only a subset of the database that is of particular interest to a user and provide a level of security by restricting access to certain data.

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2) What is a data model? Discuss the main types of data model.

A data model is an integrated collection of concepts used to describe and manipulate data, relationships between data, and constraints on the data. It represents real-world objects and their associations.

The main types of data models are:

- **Object-based data models**, such as the Entity-Relationship and Object-Oriented models.
- **Record-based data models**, including the relational, network, and hierarchical models.
- **Physical data models**, which describe how data is stored, including record structures and access paths.

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3) Discuss each of the following terms:

a) Entity – A real-world object represented in the database, typically as a table.
b) Attribute – A named column of a relation.
c) Relationship – An association between entities.
d) Domain – The set of allowable values for an attribute.
e) Tuple – A row in a relation.
f) Degree – The number of attributes in a relation.
g) Cardinality – The number of tuples in a relation.
h) Relational Database Schema – A set of relation schemas, each with a distinct name.
i) Relational Keys – Attributes that uniquely identify each tuple in a relation.

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4) What are integrity constrains? Discuss entity and referential integrity.

Integrity constraints are rules that restrict the values that can be stored in the database to ensure accuracy and consistency.

Entity integrity requires that no attribute of a primary key can be null.
Referential integrity requires that a foreign key value must either match a candidate key value in the referenced relation or be wholly null.

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1) Describe the main components in a DBMS.

The main DBMS components are the query processor, database manager, file manager, DDL compiler, DML preprocessor, and catalog manager. Together they manage query processing, storage, metadata, and access control.

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2) Describe the types of facility you would expect to be provided in a multi-user DBMS.

Facilities include data storage, retrieval and update services, authorization services, transaction support, integrity services, concurrency control, recovery services, data communication support, and utility services.

---

3) Discuss the differences between DDL and DML? What operations would you typically expect to be available in each language?

DDL is used to define and modify the database schema and constraints, while DML is used to insert, update, delete, and retrieve data from the database.

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4) Explain the following in terms of providing security for a database:

a) Data Administrator – Manages data policies, standards, and conceptual/logical design.
b) Database Administrator – Manages physical database design, security, performance, and maintenance.
c) Logical Database Designer – Identifies data, relationships, and constraints.
d) Physical Database Designer – Decides storage structures, access methods, and security measures.
e) Application Developer – Develops programs that access the database.
f) End-Users – Use the database to satisfy information needs.

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5) To address the issue of data independence, the ANSI-SPARC three-level architecture was proposed. Compare and contrast the three levels of this model.

The external level represents user views, the conceptual level represents the logical structure of the database, and the internal level represents physical storage. The separation provides data independence.

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6) What is a view? Discuss the difference between a view and a base relation.

A view is a virtual relation defined by a query on base relations. Unlike base relations, views do not store data physically.

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7) Outline the advantages and disadvantages of a DBMS.

Advantages include reduced redundancy, improved consistency, data sharing, and security. Disadvantages include complexity, cost, performance overhead, and higher impact of failures.

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1) Describe the major components of an information system.

The major components are the database, database software, application software, hardware, and people.

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2) Discuss the relationship between the information systems lifecycle and the database system development lifecycle.

The database lifecycle is part of the broader information systems lifecycle, supporting organizational requirements through planning, design, implementation, testing, and maintenance.

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3) Describe the main purpose(s) and activities associated with each stage of the database system development lifecycle.

Stages include database planning, system definition, requirements collection, database design, DBMS selection, application design, prototyping, implementation, data conversion, testing, and operational maintenance.

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4) Discuss what a user view represents in the context of a database system.

A user view defines the data and transactions required for a specific role or application area.

---

5) Discuss the main approaches for managing the design of a database system that has multiple user views.

Approaches include centralized design, view integration, and a combination of both.

---

6) What are the criteria to produce an optimal data model?

Criteria include structural validity, simplicity, expressibility, nonredundancy, shareability, extensibility, integrity, and diagrammatic representation.

---

7) Compare and contrast the three phases of database design.

Conceptual design is DBMS-independent, logical design maps data into a logical model, and physical design focuses on storage and performance.

---

1) Describe the main objective of conceptual database design.

To build a conceptual data model representing enterprise data requirements using entities, relationships, attributes, keys, and constraints.

---

2) Identify the main steps associated with conceptual database design.

Steps include identifying entities, relationships, attributes, domains, keys, checking redundancy, validating transactions, and reviewing with users.

---

3) Identify the stage(s) where it is appropriate to select a DBMS and describe an approach to selecting the 'best' DBMS.

DBMS selection typically occurs after conceptual and logical design, using evaluation and comparison of shortlisted systems.

---

4) Application design involves transaction design and user interface design. Describe the purpose and main activities associated with each.

Transaction design defines database operations, while UI design focuses on usability, consistency, and error handling.

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5) Discuss why testing cannot show the absence of faults, only that software faults are present.

Testing can reveal errors but cannot guarantee the absence of faults; it only demonstrates correct behavior under tested conditions.

---

6) Describe the main advantages of using the prototyping approach when building a database system.

Prototyping allows early user feedback, is inexpensive, quick to build, and helps refine requirements.

---

1) Describe the purpose of normalizing data.

Normalization reduces redundancy, improves integrity, and supports efficient data maintenance.

---

2) Describe the types of update anomalies that may occur on a relation that has redundant data.

Insertion, deletion, and modification anomalies.

---

3) Describe the concept of functional dependency.

A functional dependency A → B exists when each value of A determines exactly one value of B.

---

4) Describe how a database designer typically identifies the set of functional dependencies associated with a relation.

Through user requirements, documentation, domain knowledge, and analysis of sample data.

---

5) Describe the characteristics of a table in Unnormalized Form (UNF) and describe how such a table is converted to a First Normal Form (1NF) relation.

UNF contains repeating groups. Conversion to 1NF removes repeating groups and ensures atomic values.

---

6) What is the minimal normal form that a relation must satisfy? Provide a definition for this normal form.

First Normal Form, where each attribute contains a single atomic value.

---

7) Discuss how the definitions of 2NF and 3NF based on primary keys differ from the general definitions of 2NF and 3NF.

General definitions consider all candidate keys, not only the primary key, making normalization stricter.

---

8) Suppose that we decompose the schema R = (A, B, C, D, E)...

The decomposition is lossless because the common attribute functionally determines the relations.

---

9) Examine the Patient Medication Form...

(a) Functional dependencies are identified from attributes.
(b) Normalization proceeds from 1NF to 3NF.
(c) Primary, alternate, and foreign keys are identified accordingly.

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10) The table shown in Figure 5.2...

(a) Insertion, deletion, and update anomalies exist.
(b) Functional dependencies are identified with stated assumptions.
(c) Table is normalized to 3NF with appropriate keys.

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11) An agency called Instant Cover...

(a) Update anomalies are demonstrated.
(b) Functional dependencies are identified.
(c) Relations are normalized to 3NF.

1) Explain the purpose and scope of database security.

Database security is the set of mechanisms that protect the database against intentional or accidental threats. Its purpose is to protect the data; however, its scope extends beyond the DBMS itself and includes the entire database environment, such as hardware, software, data, users, and procedures.

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2) List the main types of threat that could affect a database system and for each describe the controls that you would use to counteract each of them.

The main types of threats are theft and fraud, loss of confidentiality, loss of privacy, loss of integrity, and loss of availability. These threats can be countered using authentication and authorization mechanisms, access control, encryption, integrity constraints, backup procedures, and recovery mechanisms.

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3) Discuss the roles of the following personnel in the database environment:

a) Authorization
The granting of rights or privileges that enable a user to access database objects.

b) Authentication
A mechanism that verifies the identity of a user attempting to access the database.

c) Access
Defines what operations a user can perform on database objects such as reading, writing, or modifying data.

d) Views
Virtual relations that present selected data to users and restrict access to sensitive information.

e) Backup and recovery
The process of creating copies of the database and log files to allow recovery in case of failure.

f) Integrity
Constraints that ensure data remains valid, consistent, and correct.

g) Encryption
The encoding of data to prevent unauthorized access without a decryption key.

h) RAID technology
An arrangement of multiple disks to improve reliability and performance.

---

4) Discuss Discretionary Access Control and Mandatory Access Control.

Discretionary Access Control (DAC) restricts access based on the identity of users and groups.
Mandatory Access Control (MAC) is based on system-wide security policies where users and data objects are assigned security levels, and access is controlled by the DBMS.

---

5) Describe the approaches for securing DBMSs on the Web.

Approaches include firewalls, proxy servers, digital certificates, message digest algorithms, SSL, Kerberos, and secure communication protocols.

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6) Describe any specific security measures for databases in mobile applications and devices.

Security measures include protection against mobile malware, encryption of data in transit and at rest, secure authentication, device access control, and protection against device loss or theft.

---

1) Explain what is meant by a transaction. Why are transactions important units of operation in a DBMS?

A transaction is a sequence of database operations treated as a single logical unit of work. Transactions are important because they ensure data consistency and reliability in multi-user environments.

---

2) The consistency and reliability aspects of transactions are due to the "ACIDity" properties of transactions. Discuss each of these properties and how they relate to the concurrency control and recovery mechanisms. Give examples to illustrate your answer.

Atomicity ensures that all operations of a transaction are completed or none are.
Consistency ensures that the database moves from one valid state to another.

Isolation ensures that concurrent transactions do not interfere with each other.
Durability ensures that committed changes persist even after failures.

---

3) Describe, with examples, the types of problem that can occur in a multi-user environment when concurrent access to the database is allowed.

Problems include lost updates, uncommitted dependency (dirty read), and inconsistent analysis.

---

4) Explain the concepts of serial, nonserial, and serializable schedules. State the rules for equivalence of schedules.

A serial schedule executes transactions one after another.
A nonserial schedule interleaves operations.
A serializable schedule produces the same result as some serial schedule.
Schedules are equivalent if they preserve the order of conflicting operations.

---

5) Discuss the difference between conflict serializability and view serializability.

Conflict serializability is based on swapping non-conflicting operations, while view serializability considers whether transactions see the same data values.

---

1) Discuss the types of problem that can occur with locking-based mechanisms for concurrency control and the actions that can be taken by a DBMS to prevent them.

Problems include deadlock, livelock, and starvation. A DBMS can prevent them using deadlock detection, timeout mechanisms, and priority schemes.

---

2) Why would two-phase locking not be an appropriate concurrency control scheme for indexes? Discuss a more appropriate locking scheme for tree-based indexes.

Two-phase locking is too restrictive for indexes due to frequent structural changes. Tree-based protocols such as B-tree locking allow higher concurrency by locking only affected nodes.

---

3) What is a timestamp? How do timestamp-based protocols for concurrency control differ from locking based protocols?

A timestamp is a unique identifier assigned to a transaction. Timestamp-based protocols order transactions by timestamps rather than locks, avoiding deadlocks.

---

4) Describe the basic timestamp ordering protocol for concurrency control.

Transactions are ordered by timestamps, and operations are allowed only if they respect this order. Violations cause transactions to be rolled back.

---

5) Describe how versions can be used to increase concurrency.

Multiple versions of data items are maintained so that transactions can access appropriate versions without blocking.

---

6) Discuss the difference between pessimistic and optimistic concurrency control.

Pessimistic control assumes conflicts will occur and uses locks, while optimistic control assumes conflicts are rare and validates transactions at commit time.

---

1) Describe the main recovery techniques used in DBMSs.

Recovery techniques include deferred update, immediate update, shadow paging, and checkpointing.

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2) Explain the purpose of the log file in recovery.

The log file records all database modifications to support undo and redo operations during recovery.

---

3) What is a checkpoint and why is it important?

A checkpoint is a synchronization point that reduces recovery time by limiting the amount of log processing required.

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4) Distinguish between hot and cold backup.

Hot backup is taken while the database is running, whereas cold backup is taken when the database is shut down.

---

1) What is a Distributed Database Management System (DDBMS)?

A DDBMS manages a logically integrated database that is physically distributed across multiple sites connected by a network.

---

2) Distinguish between distributed DBMS and distributed processing.

In a distributed DBMS, data is distributed across sites, whereas in distributed processing the database is centralized.

---

3) What is a parallel DBMS?

A DBMS that uses multiple processors and disks to execute operations in parallel to improve performance.

---

4) Discuss the advantages and disadvantages of a DDBMS.

Advantages include availability, reliability, and scalability. Disadvantages include complexity, cost, and security issues.

---

5) Distinguish between homogeneous and heterogeneous distributed database systems.

Homogeneous systems use the same DBMS at all sites, while heterogeneous systems use different DBMSs.

---

6) What additional functions are required in a DDBMS compared to a centralized DBMS?

Extended communication, distributed query processing, distributed concurrency control, and distributed recovery.

---

7) What is a multidatabase system?

A system that integrates multiple autonomous databases without full schema integration.

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8) Explain fragmentation in distributed databases.

Fragmentation divides relations into smaller fragments, either horizontally or vertically.

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9) Identify the types of transparency in distributed database systems.

Distribution, transaction, performance, and DBMS transparency.

---

1) Explain briefly how table locks work in MySQL.

Table locks lock the entire table and prevent concurrent write operations while allowing concurrent reads.

---

2) Explain briefly how row locks work in MySQL.

Row locks lock individual rows and are implemented by the storage engine, allowing higher concurrency.

---

3) Describe briefly the four SQL Isolation Levels.

Read Uncommitted, Read Committed, Repeatable Read, and Serializable.

---

4) How does Multiversion Concurrency Control work in MySQL?

MVCC maintains multiple versions of data to provide consistent snapshots for transactions.

---

5) Compare the InnoDB and MyISAM storage engines of MySQL.

InnoDB supports transactions, MVCC, and recovery, while MyISAM does not.

---

6) What factors should be taken into consideration for selecting the right engine in MySQL?

Transaction support, backup needs, crash recovery, and special features.

---

7) What are the reasons for benchmarking a MySQL server, and what can be achieved through benchmarks?

Benchmarking validates assumptions, measures performance, identifies bottlenecks, and plans scalability.

---

8) What is the aim of profiling a MySQL server performance?

To identify where time is consumed and diagnose performance problems.

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9) In terms of diagnostics what are usually the causes of poor performance in terms of hardware?

Overworked resources, misconfiguration, or hardware failure.

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10) How are single queries profiles created?

Using the SHOW PROFILE command.

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11) How do you determine whether the problem is server-wide or isolated to a single query?

By monitoring system-wide performance and analyzing logs and diagnostic data.

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12) What is the aim of query optimisation and index optimisation?

To reduce execution time by eliminating unnecessary work and improving data access paths.

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1) Explain what has led to the explosion of data and identify sources of Big Data.

The explosion of data is due to mobile devices, sensors, social media, video systems, and enterprise systems.

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2) Provide a definition of Big Data.

Big Data refers to data whose volume, velocity, and variety exceed the capabilities of traditional data processing systems.

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3) Explain briefly why Big Data has become popular today.

Advances in scalable infrastructure, analytics, and reduced costs have enabled Big Data processing.

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4) Define the 4Vs and identify the Big Data characteristics.

Volume, Velocity, Variety, and Veracity.

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5) Identify the Data Processing techniques for Big Data.

Centralized and distributed data processing.

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6) Describe the two processing architectures for Big Data.

Shared-everything and shared-nothing architectures.

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7) Describe the Big Data technologies?

Hadoop, HDFS, MapReduce, HBase, ZooKeeper, and Avro.

---

8) Describe the NoSQL databases and explain why RDBMS are not suitable for Big Data?

NoSQL databases include key-value, column-family, document, and graph databases. RDBMSs do not scale efficiently for Big Data workloads.

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9) Use a diagram to define the main phases of the Big Data Lifecycle.

Data acquisition, data serialization, data aggregation, data mining, and information dissemination.